

Virtual Power Plant

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Date Generated: Tue Jul 07 2026

Executive Summary

High level system description

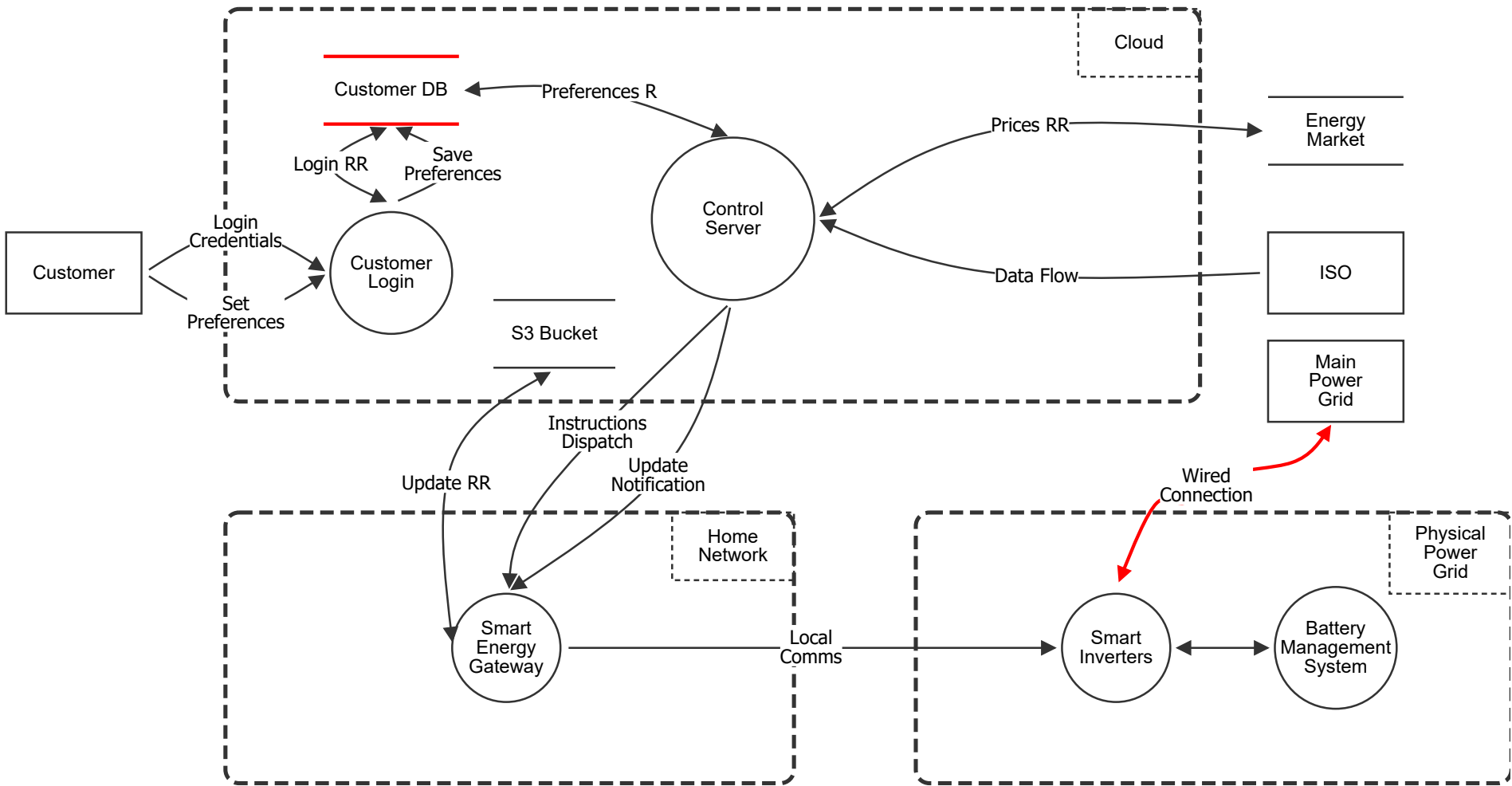
Not provided

Summary

Total Threats	10
Total Mitigated	8
Total Open	2
Open / Critical Severity	0
Open / High Severity	1
Open / Medium Severity	1
Open / Low Severity	0

STRIDE

STRIDE diagram



STRIDE

Data Flow (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
9	Jamming	Denial of service	Critical	Mitigated		Block ISO instructions from being executed. Goals: national security compromisation, compromise infrastructure functionality or reputation when more energy is requested. Boycott events that require extra energy.	Regularly ping ISO channel to confirm availability. Have backup communication channels (IP, satellite connection, cable phone).

Local Comms (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
10	Change data read by SEG	Tampering	Medium	Mitigated		Manipulate energy levels read by SEG from inverter in order to consume more energy or steal it. (Severity related to amount of compromised nodes)	Validate energy levels by monitoring energy sent and received. (Energy production from solar panels can only be estimated) Encrypt with AES-256 communications between SEG and inverter.

Prices RR (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
6	Jamming	Denial of service	Low	Mitigated		Delaying or denying updates of energy prices. Goal: sell/buy energy at more convenient prices	Ping energy market server regularly to confirm availability. Have multiple HWs (in different geographical locations) to make it harder to jam all channels.

Customer (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Impersonation	Spoofing	Critical	Mitigated		An attacker might use stolen credentials to authenticate on the portal.	Strong password requirements, MFA, prevention.

Customer DB (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
3	Data leak	Information disclosure	High	Open		Customer info disclosure, linked to: 1. Customer Spoofing (single leak) 2. Elevation of Privilege (whole DB leak)	1. Prevent Customer Spoofing (see Customer Threats) 2. Enable MFA, Hire Physical Security

Energy Market (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
8	False market prices	Tampering	Low	Mitigated		Attackers modify market prices on the Energy Market. Goals: sell/buy energy at more convenient prices.	Cross-validate prices from different authorities. Only acquire data from trusted Energy Markets.

Customer Login (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
11	Phishing	Spoofing	Critical	Mitigated		E-mail, SMS or voice call phishing attacks that lead you to a forged login page to steal your credentials or gain an active session	Provide clear instruction to users for the procedures that the company will or will not use (e.g. e-mails and SMS will never contain links, and voice calls will never be used).

ISO (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
4	Spoofed ISO instructions	Spoofing	Low	Mitigated		Impersonating ISO to send false dispatch instructions. (Threat could come from an ISO employee) Goals: Energy redirection, DoS, stealing energy, market manipulation	Double check for suspicious instructions with automated threat detections. (Generate a warning for IT department). Have limits on instructions that can be executed without human supervision, both for energy levels and frequency. Request that ISO implements MFA.

S3 Bucket (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
2	Scraping	Information disclosure	Low	Mitigated		Attackers scrape the DB and gain information about the latest firmware version.	Detect fake or compromised accounts. Information accessible only if authenticated and authorised.

Wired Connection (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
5	Wire tampering	Denial of service	Medium	Open		Cut or overheat wire.	Have SEG monitor functionality. Have dispatch service to fix broken wires.